

Power-Conjugacy Saturation for Unknot Recognition

Sharp graph certificates, polynomial deletion closure,
and binary power-pair kernels

Research contribution prepared with ChatGPT
for Vladimir Reshetnikov's *ProveIt* project

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Abstract

We develop a source-checkable group-presentation operation complementary to ProveIt's primitive-pair and unit-coordinate forest contractions. Verified relations between conjugates of generator powers define a labelled graph. Using the established balancedness property of classical knot groups, an inconsistent absolute exponent ratio around a cycle forces every generator in its connected component to be trivial. Plain power equalities admit an additional signed-cycle test. We prove that these tests are complete for the graph abstraction: every surviving component has a simultaneous nontrivial realization inside the trefoil group when the discarded conjugator labels are treated as independent ambient elements. Thus the distinction between absolute and signed consistency is necessary, not an implementation choice.

We give short independently replayable certificates, deterministic exact fallback behind a fixed-prime sieve, and a conservative polynomial bit bound for complete deletion-only saturation of literal presentations. Binary power-pair families permit polynomial-cost elimination where the existing local primitive-forest eligibility conditions provide no move; a gcd-of-minors criterion also yields an unconditional two-generator triviality theorem. A self-contained Python package includes an independent source replayer, a small reduced-Khovanov reference oracle, 24 passing test methods, and a 3227-input braid audit. The audit has no incorrect positive verdict, but adds no coverage over pure-power deletion on that corpus. Controlled sieve measurements retain both gains and arithmetic aliasing failures. These are local and restricted-input results, not an unrestricted quasi-polynomial unknot-recognition theorem or a measured speedup of the maintained recognizer.

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1 Research target and relationship to the repository

The input problem is to decide whether an explicitly encoded classical knot diagram in S^3 , with n crossings, represents the unknot. The intended asymptotic target is $2^{(\log n)^{O(1)}}$. A correct local simplifier is useful for this target only when its own bit cost, its effect on intermediate representations, and the cost of discovering the simplifications are all accounted for. A terminating sequence of presentation changes is not by itself a running-time theorem.

The repository inspection was pinned to commit

13d60da8e683ca02b11c6b301c068bda929014e7.

We read the current synthesis introduction and the primitive-projection, primitive-forest and group-certificate sections, together with the synthesis README and report 45's README [7]. The relevant maintained ideas are already substantial. A rank-two primitive-power terminal is implemented; disjoint primitive-pair quotients and overlapping unit-coordinate forests have independent replay; raw circuits and explicit normalization boundaries are distinguished; and later synthesis notes include arbitrary acyclic singleton-definition batches. Repeating the rank-two terminal would not constitute a continuation.

The present target is the restriction that the forest rule requires acyclic unit-coordinate dependencies. In a forest edge $c = p^k$, the child can be replaced by a power of its parent. More general relations such as

$$x^a = uy^bu^{-1}$$

need not solve either generator as an integral power. They may nevertheless force both generators to be trivial after comparison with other relations. A cycle can supply the missing information. We study that case without pretending that every overlapping nonunit relation is a safe forest edge.

Contribution and attribution. The knot-group balancedness lemma used below is prior work, as are Artin presentations, the cyclic-knot-group criterion and Khovanov unknot detection [1, 3, 4, 5]. The contribution here is the exact scope of the labelled-graph inference, its sharpness by a fixed-knot countermodel, its certificate and saturation interfaces, and the associated restricted complexity results. The gcd-of-minors argument is elementary and is presented with its full proof, not as an independently established literature-priority claim. No exhaustive novelty search or external referee validation is claimed.

The global target remains distinct. Lackenby's inspected 2026 preprint estimates the number of passes through a hierarchy algorithm in terms of a hierarchy-length bound L and a surface complexity bound g . Its Section 9 separates that estimate from the bit cost of the operations and gives $L(g+1)^L$ in Proposition 9.1 [2]. The present paper supplies neither the missing global hierarchy bounds nor a replacement complete quasi-polynomial search. It isolates a presentation operation whose saturation does not require a logarithmic-depth hypothesis merely to bound its own repeated execution.

2 Source hypotheses and the topological input

Throughout, a *classical knot group* means $G(K) = \pi_1(S^3 \setminus K)$ for a single classical knot K . A source-replayed presentation is a full presentation known, through reconstruction and checked changes of presentation, to present this group. It is not an arbitrary presentation accompanied by a Boolean assertion of torsion-freeness.

Lemma 2.1 (Established knot-group facts). *A classical knot group is torsion-free. If $x, u \in G(K)$ and a, b are nonzero integers with*

$$x^a = ux^bu^{-1},$$

then either $x = 1$ or $|a| = |b|$.

The second statement is Lemma 2.3 of the arXiv version of Himeno–Motegi–Teragaito [1], with the conjugator inverted to match our convention. We refer to it as *balancedness*. It is not inferred from torsion-freeness alone.

For example, consider affine maps of $\mathbb{Q}$ with $x(t) = t + 1$ and $u(t) = t/2$. Then $x = ux^2u^{-1}$ and $x \neq 1$. The subgroup generated by these maps is torsion-free: a nonidentity map with slope 2^k , $k \neq 0$, has infinite order, and a nonzero translation also has infinite order. Consequently, a purported verifier that replaces “classical knot group” by merely “torsion-free group” would be unsound for the absolute-ratio rule.

Proposition 2.2 (Cyclic group terminal). *If a source-replayed presentation of $G(K)$ has one generator and every remaining relator has exponent sum zero in that generator, then K is the unknot.*

Proof. A word in a single generator is its exponent-sum power. Thus every relator is the identity in the free cyclic group and the presented group is $\mathbb{Z}$. The boundary homomorphism $\mathbb{Z}^2 \rightarrow \pi_1(E(K)) = \mathbb{Z}$ is not injective. The Loop Theorem supplies a compressing disc for the boundary torus. The knot exterior is irreducible: a sphere in the exterior has a ball on the side not containing the connected knot. A compressible torus boundary in this irreducible exterior makes the exterior a solid torus, equivalently providing the longitude disc and hence a spanning disc for K . This is the standard cyclic-knot-group criterion [4]; the maintained group checker uses the same full-group criterion [7]. $\square$

Two invalid shortcuts. First, $G(K)^{\text{ab}} \cong \mathbb{Z}$ holds for every knot, so abelianization alone cannot authorize Proposition 2.2. Second, torsion-freeness does not imply that $v^d = w^d$ forces $v = w$. The distinction will be especially important for inverse conjugacy and for the power-pair arguments below. Those arguments compare powers of the *same* element or use explicit Bézout combinations of equal powers; they do not assume a general unique-root property.

3 The power-conjugacy graph and its sound inference rules

Definition 3.1 (Verified power-conjugacy edge). Let $X = \{x_v : v \in V\}$ be current live generators. An oriented edge $e : s \rightarrow t$ has nonzero signed integers (a_e, b_e) and a checked relation

$$x_s^{a_e} = u_e x_t^{b_e} u_e^{-1}. \tag{1}$$

It is *plain* only if $u_e = 1$ is established. The supplied implementation uses the literal empty word for this flag. The underlying graph is undirected for connectivity purposes and permits loops and parallel edges.

An inverse traversal interchanges (a_e, b_e) and uses u_e^{-1} as its conjugator. A relator

$$(x_s^a u x_t^{-b} u^{-1})^d = 1, \quad d > 0, \tag{2}$$

provides an edge by torsion-freeness. The source replayer must check the relator, the repeated-root decomposition and all cuts before using this consequence. The graph routine itself has no access to the knot source.

Lemma 3.2 (Path composition). *For a directed traversal P with consecutive exponent pairs $(a_1, b_1), \dots, (a_k, b_k)$, write $A_P = \prod_i a_i$, $B_P = \prod_i b_i$, and let U_P be the product of the traversed conjugators in path order. Then*

$$x_{\text{start}(P)}^{A_P} = U_P x_{\text{end}(P)}^{B_P} U_P^{-1}. \quad (3)$$

Proof. For two successive edges, raise the first equality to a_2 and the second to b_1 . The identity $(uzu^{-1})^k = uz^k u^{-1}$, valid for every integer k , gives

$$x_0^{a_1 a_2} = u_1 x_1^{b_1 a_2} u_1^{-1} = u_1 u_2 x_2^{b_1 b_2} u_2^{-1} u_1^{-1}.$$

Induction proves the statement. No extraction of roots and no multiplication of noncommuting base elements is used. In particular, the conjugators are not exponentiated when paths are composed. $\square$

Lemma 3.3 (Propagation of triviality). *For a verified edge in a torsion-free group, $x_s = 1$ if and only if $x_t = 1$.*

Proof. If $x_s = 1$, Equation (1) implies $x_t^{b_e} = 1$; since $b_e \neq 0$, torsion-freeness gives $x_t = 1$. Reverse the edge for the converse. $\square$

Definition 3.4 (Two consistency conditions). A connected component C is *absolutely consistent* if there are $q_v \in \mathbb{Q}_{>0}$ satisfying

$$|a_e|q_s = |b_e|q_t \quad (e \in E(C)). \quad (4)$$

Its plain-edge subgraph is *sign-consistent* if there are $\epsilon_v \in \{1, -1\}$ satisfying

$$\epsilon_t = \text{sign}(a_e b_e) \epsilon_s \quad (e \text{ plain}). \quad (5)$$

The choices in different connected components of the plain subgraph are independent. A full component is *bad* if absolute consistency fails or plain sign consistency fails. Let $\mathcal{D}$ be the union of its vertices over all bad components.

Theorem 3.5 (Sound graph annihilation). *For edges verified in a source-replayed classical knot group, every x_v with $v \in \mathcal{D}$ is the identity. Adjoining these identities, deleting the corresponding generators, and substituting the identity for them in every retained relator preserves the presented knot group.*

Proof. Choose a spanning tree of a full component. Assign $q = 1$ at its root and propagate the rational ratios $|a|/|b|$ along tree edges. A failed nontree edge gives a fundamental cycle P with $|A_P| \neq |B_P|$. Conversely, if all such edges agree, the propagated values satisfy every edge. Thus absolute inconsistency has a simple-cycle witness. By Lemma 3.2, its base generator has conjugate powers with different absolute exponents and is trivial by Lemma 2.1. Lemma 3.3 kills the entire component.

If the absolute test succeeds but the plain sign test fails, repeat the tree argument on a connected component of the plain subgraph. It supplies a plain cycle with $A_P = -B_P \neq 0$. Its conjugator is the identity, so $x^{A_P - B_P} = 1$; torsion-freeness gives $x = 1$. Propagation again uses all edges, not just the plain ones.

Each resulting identity already holds in the full presented group. Adjoining it therefore changes no quotient. Tietze elimination of the identity generators means deleting their letters from *all* relators. Other relators are not discarded simply because they were not selected as donors. The group remains isomorphic to the same source knot group, so the hypotheses are retained for the next pass. $\square$

Elementary examples. The plain relations $x^2 = y^3$ and $x^5 = y^7$ give a two-edge cycle with products 14 and 15, so $x = y = 1$. The same conclusion holds if the two equalities have unrelated conjugators, provided the ambient group is a classical knot group. In contrast, $x = ux^{-1}u^{-1}$ is not a valid annihilation certificate for arbitrary knot groups. The next section shows that this last distinction is optimal for the graph abstraction.

4 Sharpness: every survivor is realizable in one fixed knot group

The graph drops most source information. In particular, it forgets the words spelling the u_e , dependencies among those words, peripheral data, and the requirement that the named vertices generate the entire group. Sharpness must be stated for that abstraction, not for the original presentation.

Definition 4.1 (Abstract knot-model). An abstract knot-model of a labelled graph is a classical knot group $G(K)$, assigned elements $x_v \in G(K)$, and independently chosen ambient elements $u_e \in G(K)$ satisfying all edge relations. Plain edges have $u_e = 1$. There is no assertion that the x_v generate $G(K)$ or that a specified source word evaluates to the chosen u_e .

Theorem 4.2 (Exact forced-identity set). *A vertex is the identity in every abstract knot-model if and only if it belongs to $\mathcal{D}$. Moreover, a single model in the trefoil group makes all vertices outside $\mathcal{D}$ simultaneously nontrivial.*

Proof. The “if” assertion is Theorem 3.5. For the converse use the fixed group

$$T = \langle a, b \mid a^2 = b^3 \rangle$$

of the trefoil. Put $g = [a, b] = a^{-1}b^{-1}ab$. The common element $a^2 = b^3$ is central, and the commutator identity gives

$$1 = [a^2, b] = (a^{-1}ga)g.$$

Hence $a^{-1}ga = g^{-1}$; conjugation by a also inverts g because a^2 is central. The quotient $T \rightarrow C_2 * C_3$ sends g to the nonempty reduced alternating word $ab^{-1}ab$, so $g \neq 1$. Torsion-freeness of T makes g infinite-order. This is the trefoil specialization of the standard even-torus-knot example in [1]; the argument is included to specify exactly what the construction uses.

For each good component, choose positive rational potentials q_v from Equation (4), clear a common denominator, and obtain positive integers n_v . Choose signs ϵ_v satisfying the plain constraints. Set

$$x_v = g^{\epsilon_v n_v}.$$

Every such element is nontrivial. For an edge let $A = a_e \epsilon_s n_s$ and $B = b_e \epsilon_t n_t$. Absolute consistency says $|A| = |B|$. If $A = B$, choose $u_e = 1$; if $A = -B$, choose $u_e = a$. In either case $g^A = u_e g^B u_e^{-1}$. For a plain edge, Equation (5) ensures $A = B$, so its conjugator is indeed the identity. On every bad component assign all vertex elements to 1 and all conjugators to 1. Different full components have no edges between them, completing the simultaneous model. $\square$

Corollary 4.3 (A precise algorithmic barrier). *No sound rule using only the graph’s endpoint, exponent and plain-flag data can universally delete an additional surviving vertex in classical knot groups. More coverage requires additional source information or a change of presentation, not a more aggressive interpretation of these same labels.*

This is a barrier for the *abstraction*, not a lower bound for unknot recognition. The trefoil assignment may violate the actual source conjugator words or its abelianization marking. The original presentation can therefore contain decisive information that this graph deliberately forgets.

A topologically conditional strengthening. In a knot group having no nontrivial element conjugate to its inverse, any signed inconsistency around a full graph cycle is sufficient: a relation $x^A = ux^{-A}u^{-1}$ makes the nonidentity element x^A conjugate to its inverse. For example, the hyperbolic-knot case has this property by [1, Corollary 1.8]. The source must establish the stronger hypothesis; the implementation does not assume hyperbolicity. A generic “no torsion” flag cannot replace it.

5 Algorithms, short certificates and bit cost

5.1 Exact discovery without factoring exponents

The basic algorithm first forms full connected components. Within each, a breadth-first search propagates positive fractions using $|a|/|b|$. At an inconsistent edge, the two paths to the tree root and that edge supply a simple cycle. The whole component is marked for deletion and its remaining rational queries are skipped. Good components receive a second search in the plain subgraph propagating signs. There is no integer factorization and no expansion of a conjugator.

Let $r = |V|$, $m = |E|$, and let $B \geq 1$ bound the bit length of each absolute edge exponent. A simple tree path has at most $r - 1$ edges. A numerator or denominator formed before cancellation therefore has $O(rB)$ bits. Fraction reduction cannot increase that bound. Products used in a fundamental-cycle comparison have the same order of bit size.

Proposition 5.1 (Conservative graph bound). *The exact inference has an implementation with*

$$O\left((r + m) [rB + \log(r + m + 2)]^2\right) \quad (6)$$

bit operations, using ordinary quadratic integer arithmetic. A conservative storage bound is $O(r^2B + mB + (r + m) \log(r + m + 2))$ bits. Exponent parsing is included. Conjugator verification is not included; it is an input obligation.

Proof. There are $O(r + m)$ graph visits and fraction updates. Each update consists of a bounded number of multiplications, divisions, gcd computations and comparisons on $O(rB)$ -bit integers. Classical quadratic bit bounds suffice. Dense relabelling and array adjacency avoid requiring an unproved hashing hypothesis; sorting/comparison-map overhead is absorbed by the displayed bound. Store at most one fraction and predecessor record per vertex and one pair of exponents per edge. The sign pass is less expensive. $\square$

The Python reference uses dictionaries, lists and **Fraction**. It is an exact implementation of this arithmetic algorithm, not a claim that Python dictionary operations or big-integer operations cost one bit step. The displayed bound is deliberately conservative; improving it is not necessary for the subsequent polynomial saturation theorem.

5.2 Independent replay

A local certificate contains a sorted deleted-vertex list and one cycle witness per bad full component. A cycle is a list of edge identifiers and directions, together with the mode **modulus** or **plain**. It contains no claimed products that an attacker can inflate or falsify. The checker validates the walk, its closure and simplicity, recomputes signed products, checks either $|A| \neq |B|$ or $A \neq B$ with every edge plain, and independently reconstructs full connectivity by union-find. It requires the deleted list to equal exactly the union of the justified components.

Distinct certified components are vertex-disjoint. Taking one simple cycle per component uses at most r edge occurrences in total (including loops and two-edge parallel cycles). Thus

the combinatorial witness has $O(r \log(r + m + 2))$ bits, apart from the source edge data already supplied. Balanced product trees check all numeric cycle defects in $O(M(rB) \log(r + 2))$ bit operations under the usual regular multiplication cost assumptions; ordinary multiplication gives a polynomial bound without those assumptions. Source reconstruction, edge authentication and the connectivity scan must still be charged separately.

The checker intentionally does not rerun fraction discovery or compare its answer to the producer's answer. It verifies a shorter mathematical consequence. A proof with no deleted vertices is a harmless local noncertificate; the source saturation replayer requires a nonempty deletion in every recorded round.

5.3 A deterministic finite-field sieve

Before rational discovery, the optional sieve repeats ratio propagation in $\mathbb{F}_p^\times$, with $p = 65537$. It skips an edge if either exponent vanishes modulo p . A modular inconsistency gives a cycle whose exact integer products cannot have equal absolute values. The ordinary exact cycle checker still replays that witness. Every remaining full component then receives the rational and plain-sign passes.

Proposition 5.2 (One-sided sieve with complete fallback). *The sieve-plus-fallback algorithm returns exactly $\mathcal{D}$ for every input. It has no probabilistic equality assumption. Its worst-case bound is not better than Proposition 5.1, because an inconsistent integer cycle may be consistent modulo p .*

Proof. An edge used by the field search has invertible numerator and denominator. If its discovered cycle had $|A| = |B|$ in $\mathbb{Z}$, their reductions would be equal in the field, contradicting the discovery. Skipped edges cannot make a produced cycle false. A missed contradiction is processed by the exact pass. The sign pass has the same role as before. $\square$

Here the failure is particularly transparent. For b divisible by 32,

$$\frac{2^b + 3}{2^b + 1} \equiv 2 \pmod{65537},$$

since $2^{16} = -1$ and $2^{32} = 1$ in the field. The element 2 has order 32. A cycle of length r with that ratio on every edge is inconsistent over $\mathbb{Q}$, but invisible to this sieve whenever $32 \mid r$. A field pass can therefore add work on a whole natural family of power-of-two-sized benchmarks. Section 9 retains precisely this control rather than treating modular agreement as an exact result.

6 Exposed donors and polynomial deletion-only saturation

6.1 Literal source extraction

Let the current presentation have r live generators and fixed relator slots $R_1, \dots, R_h$. Write

$$N = r + h + \sum_i |R_i|$$

for its literal size in letters and slots, with generator identifiers renumbered into $\{1, \dots, r\}$ for complexity accounting.

The extractor first freely and cyclically reduces each relator and finds its shortest repeated literal root by the standard prefix-function period test. A uniform nonempty relator is a single-generator power; in a knot group it proves that generator trivial and is represented by a plain

self-edge $x^2 = x$. Otherwise, for each cyclic signed-run boundary in a root of length ℓ , it tests exact decompositions

$$x^a U y^{-b} U^{-1}, \quad a, b \neq 0, \quad (7)$$

with a maximal initial uniform signed run. The length $k = |U|$ determines the middle run length $\ell - |a| - 2k$.

For a fixed rotation, precompute the ends of uniform runs and the longest common prefix of the candidate U region with the inverse of the whole rotated word. The latter says exactly which k make the suffix equal to U^{-1} . Each possible k then needs a constant number of indexed checks. There are $O(\ell)$ rotations and $O(\ell)$ choices per rotation, so the scan and its number of candidate outputs are $O(\ell^2)$ at the letter-operation level. Identical edge data are deduplicated. We do not claim to find every algebraically implied relation of the form (7); the claim is exhaustive discovery in this specified literal exposed class.

A donor proof names its source slot, a root period, a rotation, the head length and the conjugator length. The separate decoder checks periodicity, uniform runs and inverse suffixes directly against the current source relator. It does not use the producer's prefix-function or longest-prefix routine. A nonminimal but valid repeated-root period would still be sound; minimality is an extraction optimization, not a trust requirement.

6.2 All deletion passes have a polynomial bound

The saturation loop is: extract donors, infer and replay the killed set, delete its letters from every relator, freely/cyclically reduce, and repeat. A nonempty pass deletes at least one generator. An empty pass ends the loop. No generator is introduced, no relator slot is silently removed, and no word length increases.

Theorem 6.1 (Polynomial literal saturation). *On a supplied source-replayed literal presentation of size N , complete saturation of this exposed donor class terminates after at most r nonempty deletion passes and one final unsuccessful pass. Discovery and independent replay have a conservative bound*

$$O(N^5 \log^2(N + 2)) \quad (8)$$

in bit operations. A polynomial-size trace suffices. No logarithmic round-depth hypothesis is needed for this deletion-only closure.

Proof. Deletion followed by reduction never increases the literal size. Every round has at most $O(N^2)$ deduplicated candidate edges, each exponent has $O(\log(N + 2))$ bits, and there are at most N live generators. Proposition 5.1 is bounded by $O(N^4 \log^2(N + 2))$ per pass. Independent donor replay can scan a complete relator for each of $O(N^2)$ candidates, costing at most $O(N^3 \log(N + 2))$ bits per pass; the direct implementation is intentionally allowed this redundancy. Word deletion and cyclic reduction are smaller. Even naive worst-case duplicate comparisons remain polynomial and fit the coarse per-pass bound. Multiply by at most $N + 1$ passes. Each donor record has a bounded number of $O(\log N)$ -bit fields, so retaining all round records is polynomial as well. Soundness is preserved by Theorem 3.5 at each pass. $\square$

The theorem concerns an already supplied literal presentation. The braid-to-Artin-word construction can have exponentially long words before it reaches this stage, and the implementation caps that construction. Neither the cap nor the polynomial closure theorem converts that frontend into a complete fast general recognizer.

6.3 What survives under compressed deletion

Let a raw straight-line program contain S rules and represent the current relators, with maximum represented-length bit size B . For a deleted set D , the homomorphism $x \mapsto 1$ for $x \in D$ and $x \mapsto x$ otherwise is evaluated by one bottom-up pass. It creates at most one image node per original rule, plus terminals and the empty node. Lengths cannot increase. Re-evaluating an immutable original grammar using the cumulative deletion mask avoids counting old allocated nodes as active new source size.

Proposition 6.2 (Conditional compressed closure). *Suppose a source-checked donor interface can completely inspect these raw masked grammars in cost $P(S, B, r, h)$, producing at most m edges with B_e -bit exponents and verifying their source witnesses within the same bound. Then at most $r + 1$ such calls, r linear grammar mappings and $r + 1$ applications of Proposition 5.1 suffice for deletion closure. Active raw grammar size and represented-length bit size do not multiply from one deletion pass to the next.*

This statement does not provide the assumed donor-discovery interface. A raw masked grammar can contain cancellations; discovering all the exposed relations after compressed normalization has a separate cost. The delivered graph kernel supports binary exponent data, and the literal source wrapper is implemented, but a complete native WordArena donor collector is not. The proposition records the useful nonexpansion invariant without hiding that missing step.

7 Binary power blocks beyond forest eligibility

7.1 An unconditional gcd-of-minors theorem

Consider a two-generator group presented solely by plain power equalities

$$H = \langle x, y \mid x^{a_i} = y^{b_i}, \quad 1 \leq i \leq m \rangle, \quad a_i, b_i \in \mathbb{Z} \setminus \{0\}. \quad (9)$$

Set $D = \gcd_{i < j} |a_i b_j - a_j b_i|$, taking the gcd of an all-zero family to be zero.

Theorem 7.1 (Exact triviality criterion for two-power blocks). *The group in Equation (9) is trivial if and only if $D = 1$. The implication $D = 1 \Rightarrow x = y = 1$ remains valid when these relations are a selected subset of a larger presentation, without torsion-freeness or any knot-group hypothesis.*

Proof. For every i, j , comparison of equal powers gives

$$x^{a_i b_j} = y^{b_i b_j} = x^{a_j b_i}, \quad y^{b_i a_j} = x^{a_i a_j} = y^{b_j a_i}.$$

Thus both x and y have every nonzero minor as an annihilating exponent. A Bézout combination of those integers gives $x^D = y^D = 1$. In particular, $D = 1$ makes both elements trivial, in any ambient group.

Conversely, the abelianization of H is the quotient of $\mathbb{Z}^2$ by the lattice spanned by $(a_i, -b_i)$. If this lattice has rank less than two, that quotient has positive rank and H is nontrivial. At rank two its index is the gcd of the absolute 2×2 minors, namely D ; this is the elementary rank-two Smith-normal-form index formula. If $D > 1$, the abelianization is again nontrivial. These alternatives exhaust $D \neq 1$. $\square$

A certificate need not contain a unit minor. For the rows $(2, 3), (4, 9), (5, 4)$, the absolute minors are 6, 7, 29; none is one, but their gcd is one. The delivered checker recomputes only a

selected list of source-row minors and checks their gcd. During production, after the first nonzero minor, every retained new minor strictly lowers the positive gcd by a factor of at least two. For B -bit coefficients this needs only $O(B)$ retained minor pairs. Scanning all pairs costs $O(m^2 B^2)$ with ordinary arithmetic; the selected index witness has $O(B \log(m+2))$ bits and polynomial replay cost. The checker accepts no unverified claimed determinant or Bézout coefficient.

This result does not say that abelianization detects triviality for arbitrary two-generator presentations. The pure two-power shape is the entire reason each lattice minor annihilates each individual generator. Likewise, $D > 1$ does not by itself describe the nonabelian structure of H .

Proposition 7.2 (Collinear rows). *If all rows in Equation (9) are integer multiples $k_i(a, b)$ of a primitive integer vector (a, b) , and $d = \gcd_i |k_i|$, then the presentation is equivalent to the single relation $x^{ad} = y^{bd}$.*

Proof. Put $u = x^a$, $v = y^b$. The given relations say $u^{k_i} = v^{k_i}$. Choose $\sum c_i k_i = d$. Multiplying the equalities after taking powers c_i gives $u^d = v^d$: the left factors are powers of u , and the right factors are powers of v , so each side can be combined independently. Conversely, $d \mid k_i$ implies every original equality from $u^d = v^d$. No unique-root property has been used. $\square$

7.2 A binary family that a unit forest cannot contract

For $M \geq 2$ and $k \geq 1$, define

$$P_{k,M} = \left\langle t, x_1, y_1, \dots, x_k, y_k \mid x_i^M = y_i^{M+1}, \quad x_i^{M+1} = y_i^{M+2} \quad (1 \leq i \leq k) \right\rangle. \quad (10)$$

Each block has determinant

$$M(M+2) - (M+1)^2 = -1.$$

Theorem 7.1 therefore eliminates every x_i, y_i in an arbitrary group and leaves the free generator t . In particular, $P_{k,M} \cong \mathbb{Z}$ without using external knot provenance.

The word $x^M y^{-(M+1)}$ is not primitive in $F(x, y)$: its one-relator quotient subjects onto the nonabelian free product $C_M * C_{M+1}$, whereas killing a primitive word would give a cyclic quotient. The same argument applies to $x^{M+1} y^{-(M+2)}$. Both have coprime exponent coordinates, so neither is a proper power. There are no singleton generator occurrences and no eligible primitive unit-coordinate donors in these displayed relators. Thus the existing *local donor conditions*, applied to this unchanged presentation, provide no forest move. A two-edge graph cycle eliminates each pair immediately.

This is a strict separation of eligible operations, not a lower bound on Whitehead search, Tietze transformations, or the maintained recognizer. The family is a supplied presentation of the unknot group; it is not being passed off as a hard family of input knot diagrams.

When $M = 2^b$, the expanded relator list has exactly $k(4 \cdot 2^b + 4)$ letters, while its two-power encoding has $O(k(b + \log(k+2)))$ bits. Direct source-row authentication and the unit-minor checker run in $O(kb^2)$ arithmetic bit cost plus indexing, without expansion. The graph kernel also remains polynomial in that binary size. Thus this family gives a fully implemented binary-input local result, not merely a claim that an unspecified compressed extractor would be efficient.

8 A source-bound braid frontend and an independent reference oracle

8.1 Reconstructing the entire group

For a braid with s strands, the closure is a knot exactly when its strand permutation is a single cycle. The frontend checks strict integer input and this condition; too few crossings to connect

all strands is rejected before a large permutation allocation. Artin meridian images are then constructed with an explicit letter allowance. At a positive crossing the current pair of meridian words (A, B) becomes

$$(ABA^{-1}, A),$$

and at a negative crossing it becomes $(B, B^{-1}AB)$. Equating every final meridian image to its initial meridian gives the full closed-braid group presentation. This is the usual Artin presentation; see [3, Theorem 1.1 and Definition 3.1].

The producer updates the tuple of current meridians in crossing order. The independent replayer instead starts with generator words and applies the corresponding literal substitutions in reverse order. This is the required reversal for pullback composition, not a discretionary change of source. All strand relations, including redundant ones, are retained. Neither replayer invokes the graph producer or the donor extractor.

8.2 The homological obstruction and a controlled basis change

Initially every Wirtinger or Artin meridian has abelianization charge one. A generator certified trivial must have charge zero. Therefore an annihilation-only rule has no direct target among unmodified meridian generators. This is a structural reason to change basis rather than merely try more graph cycles on the original meridian graph.

Choose the last meridian as t and make the explicit Nielsen change

$$x_s = t, \quad x_i = y_i t \quad (i < s).$$

The inverse is $y_i = x_i x_s^{-1}$. All y_i have charge zero and t has charge one. The substitution at most doubles the literal letter count, and both source reconstructions independently perform it. Throughout source replay, any proposal to delete t is rejected. This charge check is a necessary sanity guard, not the mathematical justification for other deletions.

After each checked round, all dead letters are removed from every relator and the surviving words are cyclically reduced. A positive certificate must end with exactly t alive and every retained relator having exponent sum zero. Proposition 2.2 then applies. Failed search returns `INCONCLUSIVE`; a construction allowance failure returns `RESOURCE_LIMIT`. Neither is a negative knot verdict.

8.3 A small exact failure of completeness

The three-braid $\sigma_1^{-1}\sigma_2$ closes to an unknot, by two successive destabilizations. In the chosen difference basis its reduced relators, writing $a = y_1$, $b = y_2$, are

$$\begin{aligned} R_1 &= ba^{-1}, \\ R_2 &= t^{-1}b^{-1}atbtb^{-1}t^{-1}a^{-1}, \\ R_3 &= t^{-1}b^{-1}atb. \end{aligned}$$

The exposed scanner finds only the two orientations of $a = b$ from R_1 . The graph is consistent, so no deletion occurs. Yet substituting $a = b$ into R_3 immediately gives $b = 1$, and then $a = 1$. The missing move is an equality contraction, already within the maintained unit-coordinate machinery. This tiny example disproves completeness of the isolated annihilation frontend, not completeness of a richer native portfolio. It also explains why integration should alternate this operation with existing non-deletion changes rather than replace them.

8.4 The Khovanov audit is a different implementation

The reference module resolves all 2^n states of a small braid diagram, constructs circles by literal connectivity, and uses the Frobenius algebra $\mathbb{F}_2[X]/(X^2)$ with

$$m(1, X) = m(X, 1) = X, \quad m(X, X) = 0, \quad \Delta(1) = 1 \otimes X + X \otimes 1, \quad \Delta(X) = X \otimes X.$$

The marked circle is fixed to label X , giving the reduced subcomplex. The code computes sparse bit-vector boundary ranks and explicitly checks $d^2 = 0$. Suppressing gradings does not change total homology dimension:

$$\dim \widetilde{\text{Kh}} = \sum_h \dim C^h - 2 \sum_h \text{rank } d_h.$$

Reduced rational homology has nonzero rank, and the universal coefficient theorem bounds it by the rank over $\mathbb{F}_2$. Together with Kronheimer–Mrowka’s unknot detection and the unknot calculation, reduced $\mathbb{F}_2$ rank one is therefore an unknot criterion [5]. The nonzero rational rank also follows directly from the normalized Jones Euler characteristic at 1.

This oracle is an exponential, capped audit backend. It is not the maintained Bar–Natan scanning implementation, whose local-complex approach is a different computational algorithm [6]. The standalone command-line option `-cube` uses this reference only after an inconclusive probe and reports a cap failure without a verdict. Tests of the new code do not constitute execution of the repository’s full maintained test suite.

9 Finite validation and measured performance

9.1 Test boundaries

The shipped standard-library-only suite has 24 passing test methods, with no failures, errors or skipped methods in the recorded run. The methods contain exhaustive and randomized cases, not merely one assertion apiece. Selected counts are in Table 1. Graph answers are compared against

Check	Count
Graph/valuation-oracle cases	3996
Exhaustive short-word donor cases	5460
Constructed donor cases	1000
Additional random homology comparisons	180
Independent Artin reconstructions	133
Random minor-gcd blocks	200
Large binary unit-pair checks	20

Table 1: Selected inner-test counts from `test_summary.json`.

a separate prime-valuation-vector oracle on small integers, in addition to comparison between exact and sieved discovery. Every positive graph answer is independently replayed. The word scanner is compared against a slower all-cut literal oracle. Other tests cover source rebinding, bad integer types, invalid cycles, forged cuts, prime-divisible exponents, field collisions, trefoil normal forms, source construction order, mirror and Markov consistency of the reference homology, large binary minors and allocation-limit behavior. These checks support implementation correctness but do not replace the mathematical proofs or amount to formal verification.

Source class	Inputs	Unknots	Certified	New*
Zero-crossing circle	1	1	1	0
Two-strand, exhaustive	170	98	98	0
Three-strand, exhaustive	2856	1712	1258	0
Four/five-strand, seeded	200	132	71	0
Total	3227	1943	1428	0

Table 2: Independent small-braid audit. “New” means positive coverage beyond pure-power-only deletion in the same difference basis, not beyond the maintained recognizer. Inputs are word presentations, not distinct knot types.

9.2 Exhaustive short-braid coverage

The corpus contains the zero-crossing circle; all signed two-braid words of lengths 1, 3, 5, 7; all single-component signed three-braid words of lengths 2, 4, 6; and 200 seeded four/five-strand knot closures with at most nine crossings. Each row is independently evaluated by the reduced homology oracle with its square-zero check enabled. All positive group certificates undergo complete source replay.

Of 3227 inputs, 1,943 have reduced rank one. The probe certifies 1,428 of them and gives no false positive. It misses 515 unknots. No source allowance is exhausted. More importantly, the pure-power-only ablation leaves exactly the same surviving generator set on every input. Every positive graph certificate seeds its deletion from an exposed single-generator power. Consequently, this corpus establishes neither new detection coverage from inconsistent nonseed cycles nor a need for the new balancedness inference on ordinary small examples. The graph can propagate a deletion in one batch rather than wait for successive powers to become exposed, but that is a different claim.

The retained example $\sigma_1^{-1}\sigma_2$ is the first missed unknot; its full relators were displayed in Section 8. It is useful negative evidence for the frontend design. The largest initial transformed presentation in the corpus has 290 letters; successful certificates use at most three deletion rounds. No extrapolation from these sizes to hard large diagrams is justified.

9.3 Fixed-prime controls and the follow-up experiment

All timed graph arms include exact independent certificate replay. Each initial case has seven shuffled repetitions of exact A/A and sieve A/A arms, with one warm-up per arm. The nine initial graph cases and twelve source cases make 588 measured calls and 84 warm-ups. Raw samples, within-round order, returned work counters, Python/platform metadata, and source hashes are retained in `benchmark.json`.

Family	r	b	Exact ms	Sieve ms	Paired ratio
unbalanced	32	64	0.622	0.932	0.605
unbalanced	128	64	2.006	2.191	0.957
unbalanced	128	256	7.663	7.747	1.021
unbalanced	512	64	18.164	18.941	0.957
balanced	32	64	0.292	0.357	0.802
balanced	128	64	1.913	2.007	0.957
balanced	128	256	7.500	7.373	1.005
prime divisible	128	64	2.590	2.600	0.910
residue collision	128	16	0.840	1.247	0.680

Table 3: Initial graph families. Exact and sieve columns are medians; “paired ratio” is the median of within-round exact/sieve time ratios. Values below one favor exact-only discovery. All times include replay.

The nominally unbalanced cycles in Table 3 have lengths divisible by 32, exactly the aliasing condition in Proposition 5.2’s discussion. Their field pass does not remove the rational work. Apparent small timing gains are not evidence of successful modular pruning; the recorded rational-edge counters and A/A variation show why. Balanced, prime-divisible and residue-collision controls likewise retain the fallback cost.

A separate mechanism-focused follow-up brackets multiples of 32 rather than replacing the initial observations. It has nine repetitions of four shuffled arms per case, for 324 measured graph calls and 36 warm-ups.

r	b	Exact ms	Sieve ms	Ratio	Exact A/A	Sieve A/A
31	64	0.288	0.154	1.769	0.992	0.895
32	64	0.315	0.351	0.838	1.019	0.925
33	64	0.309	0.165	1.906	1.052	0.982
127	256	7.499	1.460	5.525	0.968	0.981
128	256	7.771	8.578	0.934	0.884	0.942
129	256	7.348	1.373	5.480	1.008	1.028
511	64	18.750	3.340	5.758	0.992	0.973
512	64	19.102	18.905	1.000	1.030	0.954
513	64	18.723	3.342	5.546	1.033	1.001

Table 4: Cycle-length follow-up, recorded in `sieve_followup.json`. The ratio is paired exact/sieve time including replay. Adjacent multiples of 32 deliberately retain the exact fallback.

The nonmultiples produce genuine field witnesses and avoid the long rational scan; their gains are several times the A/A fluctuations. The adjacent multiples do not. These are arithmetic-family measurements, not a distribution of knot-group presentations and not a uniform asymptotic improvement. The initial and pre-validation-hardening timing records remain under `data/historical/`; current tables are generated from the current complete rerun, not from a best-of selection of samples.

9.4 Binary capacity and full reference-pipeline timings

The largest case represents $128(4 \cdot 2^{16384} + 4)$ relator letters using binary coefficients. It verifies all 256 deleted generators while retaining one isolated generator. This is a supplied-presentation capacity result. It proves neither that a hard knot diagram has this form nor that a producer can expose it cheaply.

The source timings include positive proof construction and independent source replay. The

Pairs k	b	Coefficient bits	Deleted	Median ms
1	64	65	2	0.041
16	256	257	32	0.245
128	1024	1025	256	2.791
128	16384	16385	256	27.651

Table 5: Binary family $P_{k,2^b}$. Each timed call constructs source rows, checks every unit-minor proof, builds the graph and independently replays graph deletion. Seven samples are retained per row. There is no expanded baseline completion time assigned to the large cases.

Source	n	Unknot	Cube ms	Probe+cube ms	Ratio
circle	0	yes	0.008	0.024	0.369
one crossing	1	yes	0.024	0.082	0.261
trefoil	3	no	0.101	0.169	0.605
T25	5	no	1.095	1.809	0.823
circle three	2	yes	0.075	0.164	0.421
mixed circle	4	yes	0.476	0.162	2.936
figure eight	4	no	0.330	0.574	0.667
T34	8	no	19.960	18.862	1.042
circle four	3	yes	0.251	0.228	1.126
sleeve seven	7	yes	11.032	0.363	29.305
alternating eight	8	no	16.896	19.041	0.944
five strand circle	4	yes	1.025	0.301	3.359

Table 6: Twelve fixed small sources: the same exponential reference cube alone versus the source-replayed probe followed, when needed, by that cube. These are not comparisons against **fastunknot**.

reference cube itself already costs only microseconds on the smallest examples, so a checked group probe can be slower. The seven-crossing sleeve example benefits by avoiding a full explicit cube, but contains obvious inverse crossing pairs. Its improvement is not hard-unknot evidence. The inconclusive cases pay extra work before the same cube. The A/A records remain available even where they show substantial submillisecond variation. No broad maintained-recognizer gain or change of production default is recommended on this evidence.

10 Integration contract for the maintained recognizer

The archive is an additive research package, not an installed modification of the pinned repository. Its certificate version 1 is local to the standalone braid wrapper and must not be confused with any maintained certificate-version number. The intended native handoff has four parts.

First, the maintained replayer reconstructs the caller’s original PD and replays the complete preceding presentation trace. Only that step supplies classical single-component knot provenance. Second, an exact donor adapter binds each power-conjugacy edge to a current relator slot and validates its root and conjugator cuts. The graph verifier must not accept a caller’s unproven **plain** flag or a guessed two-power interpretation of a compressed word. Third, the new graph record gives cycles and the exact component union to be deleted. Finally, the native checker substitutes identity into *every* current root and updates the live set before continuing with existing endpoints.

Policy. The small-source ablation suggests keeping this optional. Existing cheap singletons and unit-coordinate contractions should not be displaced by a more elaborate graph pass. A

promising trigger is the presence of several verified nonunit power-conjugacy edges in a component that has no already known pure-power seed. The trigger must be cheap enough not to repeat the full donor search merely to decide whether to run it. Conversely, once a useful graph exists, batch propagation can replace repeated identical connectivity work.

Safe interruption. Failure to find a witness means only that this abstraction is exhausted. Cancellation, node/work allowances and failed compressed cut verification must preserve the original valid state and return control to existing fallbacks. Deletion should be committed only after the entire batch is validated. The theorem does not authorize deleting a relator because it looked like a donor; it authorizes deleting certified identity generators and applying that substitution everywhere.

What was not run. No remote files were changed, no native WordArena adapter was installed, and the full maintained recognizer test suite was not executed for this package. The source-bound braid wrapper, its tests and its independent homology oracle are new self-contained code. The package includes an integration checklist rather than an untested production patch.

11 Further research questions and a route toward the global target

11.1 Compressed donor discovery

The most immediate algorithmic problem is to find and certify exposed forms $x^a U y^{-b} U^{-1}$ in a current compressed relator without enumerating expanded cut positions. Source equality of a proposed cut and discovery of all useful cuts are distinct tasks. Can candidate cuts be organized as arithmetic progressions, periodic families or a small set of grammar-boundary alignments? A useful theorem would bound both candidate enumeration and rejected-candidate work in the grammar size and represented-length bit size, and would remain valid after masking identity generators.

11.2 Does the cycle rule help on genuinely stalled source states?

The finite audit found no coverage beyond power deletion. The next experiment should use actual maintained search states immediately before an expensive Whitehead/overlap exposure step, not more randomly inflated circles. For every detected nonunit component, record whether a short cycle is new information, whether the old native planner already had an available contraction, and whether full source production plus replay beats the unchanged baseline. A useful outcome could be positive or negative: proving rarity would guide dispatch as much as finding gains.

11.3 Use the labels that the sharpness theorem discards

Theorem 4.2 says exactly why the graph alone has reached its limit. Retain a compressed cyclic conjugator product, prove that it is trivial or centralizes the base element, or exploit a peripheral marking. Then some cycles with opposite signed products may become safe plain-type certificates. Can such extra annotations be verified in polynomial encoded cost without requiring a general group-word solver? Any proposed extension must fail on the trefoil inverse-conjugacy example unless its extra premise genuinely excludes that model.

11.4 Geometric provenance for stronger signed inference

A verified absence of generalized torsion of order two permits signed constraints on all edges. Existing geometric characterizations in [1] provide a theorem-level premise. The algorithmic question is whether obtaining that premise is cheaper than the recognition work it saves, and how a native certificate should authenticate it. It is not legitimate to assume it merely because a diagram appears hyperbolic.

11.5 Multi-generator unconditional power blocks

Theorem 7.1 is exact for the two-generator pure-power class. What is the strongest comparably inexpensive unconditional identity rule for larger plain power graphs? Unit-defect cycles kill their own cycle vertices without torsion-freeness, but propagating to an attached vertex with nonunit exponent needs an additional argument. A generalization must not confuse a trivial abelianization with a trivial nonabelian group. Short Bézout/minor certificates and provenance-free replay are natural objects to investigate.

11.6 Amortized arithmetic and modular scheduling

The fixed-prime experiment exposes a precise failure mode, not a random hash accident. Multiple carefully chosen moduli, cheap bit-length tests on cycle products, or progress-sensitive disabling of unproductive field passes could improve constants. A deterministic implementation must still charge all moduli and preserve exact fallback. The more ambitious goal is a near-linear-in-input-bit-size weighted graph algorithm with comparably small certificates; exponent factorization should not be introduced as an uncharged oracle.

11.7 The remaining quasi-polynomial obligation

Suppose an input diagram of size n is transformed, with source replay, into polynomially many presentation states whose total encoded size and source-checked exposure cost are at most $2^{O((\log n)^c)}$. Assume useful exposures alternate with the deletion closure above until a decisive endpoint is reached. Then the displayed local polynomial bounds keep the whole alternating computation quasi-polynomial. This implication is a cost-accounting statement, not a proof of those hypotheses.

The useful improvement over a naive round-count argument is that identity-deletion phases themselves neither introduce generators nor expand the current literal words, and raw masked grammar evaluation has the analogous active-size invariant. One need not demand logarithmic *deletion* depth. The hard missing theorem concerns the availability and size of decisive exposure transformations, including those needed on knot diagrams where every graph in the current basis is consistent. A structural theorem guaranteeing such controlled exposure, rather than a faster already polynomial cycle check, is what would advance the unrestricted asymptotic target.

12 Conclusion

Power-conjugacy cycles give a valid new operation beyond the acyclic unit-coordinate forest interface. The exact graph-level boundary is now explicit: absolute inconsistencies in the full graph and signed inconsistencies in the plain subgraph generate all universally forced identities visible to this abstraction. A trefoil countermodel proves that extending the signed rule without extra hypotheses is impossible. Independent short certificates and deletion nonexpansion turn this rule into a polynomial saturation primitive, while binary power blocks provide fully checked restricted-input examples beyond the old donor eligibility.

The empirical conclusion is narrower. The delivered small-braid frontend is sound on the audited corpus but incomplete even on a two-crossing unknot, and the new nonseed cycle mechanism adds no small-corpus coverage. Modular arithmetic helps on selected bit-complexity families and fails systematically on adjacent controls. These boundaries identify the next research task: source-bound, size-controlled exposure of genuinely useful nonunit cycles in the maintained search, not a claim that the general quasi-polynomial recognition problem has been completed here.

A Certificate schema and executable examples

The graph certificate is deliberately small:

```
{"killed": [1, 2],
 "witnesses": [{"mode": "modulus", "walk": [[0, 1], [1, -1]]}]}
```

Its validity depends on the supplied authenticated edges. A direction is a strict integer $+1$ or -1 ; Booleans are not accepted as integers. A `plain` witness requires all traversed edges to be plain. A `modulus` witness is checked by exact absolute products even when field arithmetic found it. The deletion list must be sorted, duplicate-free and exactly the union of the justified connected components.

The standalone source certificate additionally contains the fixed basis change, every donor list for every round, and the terminal generator. The original braid is supplied independently to replay. A local graph proof or two-power minor proof alone is never an unknot certificate.

From the package root:

```
python code/cli.py examples/circle.json --proof /tmp/circle-proof.json
python code/cli.py examples/circle.json --verify /tmp/circle-proof.json
python code/cli.py examples/figure_eight.json --cube
python code/run_tests.py --output /tmp/power-tests.json
python code/audit_braids.py --output /tmp/power-audit.json
python code/benchmark.py --output /tmp/power-benchmark.json
python code/benchmark_sieve.py --output /tmp/power-sieve.json
python verify_manifest.py
sh build.sh
```

The last command compiles the article from LaTeX. `make_tables.py` regenerates numerical tables from the retained JSON without rerunning measurements. Re-running benchmarks gives new machine-dependent samples; the data and prose must be reviewed together before replacing the shipped observations. The tests need Python 3.10 or later and no external Python packages. Building the paper additionally needs `pdflatex` and the packages in the preamble.

B Provenance and reproducibility notes

The repository read is a pinned-source inspection, not a full checkout or a native performance baseline. The corresponding source inventory is in `integration/SOURCE_AUDIT.md`. The article's new mathematics and code were prepared for this delivery. Primary literature is attributed in the references below. Bibliographic dates should not be confused with the runtime timestamps: the research-request date is 8 October 2026, while the retained machine timestamps are UTC on 9 October 2026.

The main raw files are `data/test_summary.json`, `data/braid_audit.json`, `data/benchmark.json`, and `data/sieve_followup.json`. Each timing record includes source hashes; all measured calls and warm-ups are retained. The historical records are explicitly separate from the

current figures. Archive hashes are supplied by `MANIFEST.sha256`; it intentionally excludes itself and transient LaTeX/Python build products. The archive does not contain a Lean formalization or claim machine-checked proofs of the mathematical theorems.

References

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